

ROBOTICS AND EMBEDDED CONTROL SYSTEMS

PRACTICAL ASSIGNMENT

Giving the following definition in *Table 1*, create an Arduino sketch for a simple Ultrasonic security system that uses **Arduino uno**, **one ultrasonic sensor**, **green led**, **yellow led**, **red led**, and a **buzzer** to perform the tasks in *Table 2*. The performance of the tasks should correspond to the conditions illustrated in *Table2*.

```
#define trigPin 9 //Sensor Trig on digital pin 2

#define echoPin 8 //Sensor Echo on digital pin 3

#define LEDlampRed 7 //Red LED on digital pin 4

#define LEDlampYellow 6 //Yellow LED on digital pin 5

#define LEDlampGreen 5 //Green LED on digital pin 6

#define soundbuzzer 4 //Buzzer on digital pin 7
```

Table 1: Definitions indicating Pin location of components

Condition	Task
Sensor Distance greater than 80cm	Light the green LED
Sensor Distance greater than 50cm	Light the Yellow LED
Sensor Distance less than 20cm	Light the Red LED
Sensor distance from 0 to 20 cm	a. Sound the buzzer b. Print the distance as output

Table 2. Task required to be performed as per given conditions